21 The functions f and g are such that

$$f(x) = 3x^2 + 1 \quad \text{for } x > 0 \qquad \text{and} \qquad g(x) = \frac{4}{x^2} \quad \text{for } x > 0$$

(a) Work out gf(1)

$f(1) = 3(1)^2 + 1 = 4$

$gf(1) = g(4) = 4/4^2 = 1/4$

1 mark
f(1)=4 then g(4)

1 mark
gf(1)=1/4

1/4

(2)

The function h is such that $h = (fg)^{-1}$

(b) Find $h(x)$

$fg(x) = f(4/x^2) = 3(4/x^2)^2 + 1$

$= 48/x^4 + 1$

Let $y = 48/x^4 + 1$, so $y - 1 = 48/x^4$

$x^4 = 48/(y - 1)$, $x = \sqrt[4]{[48/(y - 1)]}$

Swap x and y: $h(x) = \sqrt[4]{[48/(x - 1)]}$, $x > 1$.

1 mark
fg(x)=$48/x^4+1$

1 mark
set y and rearrange

1 mark
isolate fourth power

1 mark
positive fourth root

(4)

(Total for Question 21 is 6 marks)

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